

Composite Solids, Surface Area, and Density Design

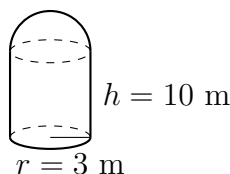
High School Geometry — Modeling with Measurement

Directions: Decide first what each solid is *made of* — which basic solids are joined, and which faces or circles disappear where they join. Then choose volume, surface area, or density. Use π from your calculator and round each answer to the precision asked for. Every figure carries only the values that problem gives you.

This review interleaves: Composite volume, composite surface area, density and mass, and reverse design problems.

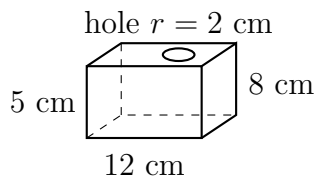
Reference: $V_{\text{cyl}} = \pi r^2 h$, $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$, $V_{\text{sphere}} = \frac{4}{3}\pi r^3$, $V_{\text{prism}} = Bh$, lateral cylinder $= 2\pi r h$, lateral cone $= \pi r \ell$, sphere surface $= 4\pi r^2$, density $= \text{mass} \div \text{volume}$.

- 1.** A grain silo is a cylinder of radius 3 m and height 10 m with a hemisphere of the same radius closing the top. Find the total volume of the silo, to the nearest hundredth of a cubic metre.



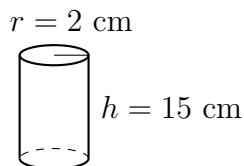
Answer: _____ m³

- 2.** A steel plate is a rectangular prism 12 cm long, 8 cm wide and 5 cm thick, with one circular hole of radius 2 cm drilled straight through its thickness. Find the total surface area of the finished plate — remember the two circles that are removed and the inside wall of the hole that is added. Round to the nearest hundredth of a square centimetre.



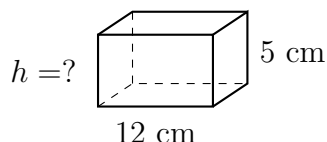
Answer: _____ cm²

3. A solid steel roller is a cylinder of radius 2 cm and height 15 cm. Steel has density 7.8 grams per cubic centimetre. Find the mass of the roller, to the nearest hundredth of a gram.



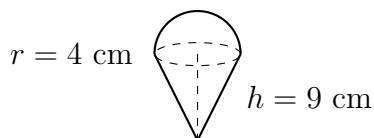
Answer: _____ g

4. A packaging designer needs a rectangular box whose base measures 12 cm by 5 cm and whose volume is exactly 900 cubic centimetres. Find the height the box must have.



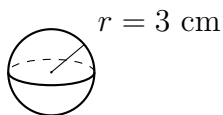
Answer: _____ cm

5. A novelty candle is a cone of radius 4 cm and height 9 cm with a hemisphere of the same radius sitting on its open top. Find the total volume of wax, to the nearest hundredth of a cubic centimetre.



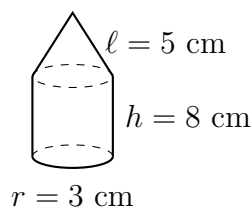
Answer: _____ cm^3

6. A solid brass ball bearing has radius 3 cm. Brass has density 8.5 grams per cubic centimetre. Find the mass of the ball, to the nearest hundredth of a gram.



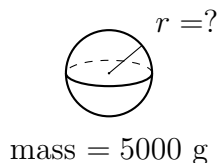
Answer: _____ g

7. A grain silo model is a cone of radius 3 cm and slant height 5 cm sitting on a closed cylinder of the same radius and height 8 cm. The circle where the cone meets the cylinder is inside the solid, so it is not painted. Find the painted outside surface area (cone slant surface + cylinder side + bottom disc), to the nearest hundredth of a square centimetre.



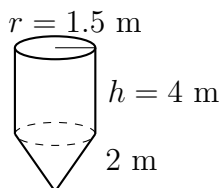
Answer: _____ cm^2

8. A sculptor is casting a solid bronze sphere and wants it to have a mass of exactly 5000 grams. Bronze has density 8.8 grams per cubic centimetre. Find the radius the sphere must have, to the nearest hundredth of a centimetre.



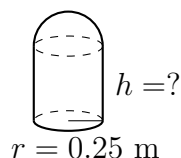
Answer: _____ cm

9. A grain hopper is a cylinder of radius 1.5 m and height 4 m with a cone of the same radius and height 2 m attached below it. Grain has density 750 kilograms per cubic metre. Find the mass of grain the hopper holds when full, to the nearest hundredth of a kilogram.



Answer: _____ kg

10. Synthesis challenge. A concrete bollard is a cylinder topped by a hemisphere of the same radius. It must contain exactly 0.30 cubic metres of concrete, and concrete has density 2400 kilograms per cubic metre.



- (a) Option A uses radius 0.25 m. Find the cylinder height h , in metres, that makes the total volume exactly 0.30 cubic metres. Round to the nearest hundredth.

Answer: _____

- (b) Find the mass of the finished bollard, in kilograms.

Answer: _____

- (c) Option B uses radius 0.30 m instead, and is also built to hold exactly 0.30 cubic metres. Explain what that forces to be true about the two bollards' masses, and give one practical reason a designer might still prefer Option B.

Answer: _____